• WHILE IMPLICITLY WRITING THE BASIC LAWS FOR A SYSTEM ARE STRAIGHTFORWARD IN THE SYSTEMS APPROACH, WE KNOW THAT WRITING THE EQUATIONS WITH A CONTROL VOLUME APPROACH IS MORE AMENABLE FOR FLUIDS. THIS MOTIVATES THE NEED FOR A THEOREM.

• SINCE A CONTROL VOLUME IS A FIXED REGION IN SPACE, INTENSIVE PROPERTIES ARE MORE AMENABLE FOR WRITING EQUATIONS FOR THE CONTROL VOLUME APPROACH.

• $N_S|_{t+\Delta t} = N_I + N_{III}|_{t+\Delta t} = N_{CV} - N_I + N_{III}|_{t+\Delta t}$ → FROM GEOMETRY, $N_S|_t = N_{CV}|_t$

$$\Rightarrow \left.\frac{dN}{dt}\right|_{SYSTEM} = \lim_{\Delta t\to 0}\frac{N_{CV}|_{t+\Delta t} - N_{CV}|_t}{\Delta t} - \underbrace{\lim_{\Delta t\to 0}\frac{N_I|_{t+\Delta t}}{\Delta t}}_{\text{NEEDS DETAILED TREATMENT.}} + \lim_{\Delta t\to 0}\frac{N_{III}|_{t+\Delta t}}{\Delta t}$$

$= \frac{\partial N_{CV}}{\partial t} = \frac{\partial}{\partial t}\int_{CV}\eta\rho\,dV$ (FOLLOWS FROM GEOMETRY DECOMPOSITION)

$dN_{III}|_{t+\Delta t} = \eta\rho\vec{v}\cdot\vec{dA}\,\Delta t \Rightarrow \lim_{\Delta t\to 0}\frac{N_{III}|_{t+\Delta t}}{\Delta t} = \int_{CS_{III}}\frac{dN_{III}|_{t+\Delta t}}{\Delta t} = \int_{CS_{III}}\eta\rho\vec{v}\cdot\vec{dA}$

* CONTROL SURFACE IS BOUNDARY OF THE CONTROL VOLUME. THE MOVEMENT OF MATERIAL ACROSS CONTROL SURFACE INLET HAS NEGATIVE SIGN, BECAUSE SURFACE AT WHICH WE DO THIS HAS OUTWARD NORMAL.

$dN_I|_{t+\Delta t} = \eta\rho\vec{v}\cdot\vec{dA}\,\Delta t \Rightarrow \lim_{\Delta t\to 0}\frac{N_I|_{t+\Delta t}}{\Delta t} = \int_{CS_I}\frac{dN_I|_{t+\Delta t}}{\Delta t} = \int_{CS_I}\eta\rho\vec{v}\cdot\vec{dA}$

ADDING, $\left.\frac{dN}{dt}\right|_{SYSTEM} = \frac{\partial}{\partial t}\int_{CV}\eta\rho\,dV + \int_{CS}\eta\rho\vec{v}\cdot\vec{dA}$

• A THING TO KEEP IN MIND IS THAT THE VELOCITY HERE HAS BEEN MEASURED W.R.T. THE CONTROL VOLUME

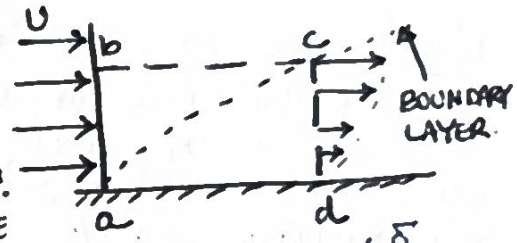
• RATE OF INCREASE OF MASS IN THE CONTROL VOLUME IS BALANCED BY THE NET FLUX OF MASS THROUGH THE CONTROL SURFACE.

• SPECIAL CASE ρ = CONSTANT $\Rightarrow \rho\frac{\partial}{\partial t}\int_{CV}dV + \rho\int_{CS}\vec{v}\cdot\vec{dA} = 0$ → NONDEFORMABLE CV $\Rightarrow \int_{CS}\vec{v}\cdot\vec{dA} = 0$ (DEFINE $Q = \int_{SURFACE} v\,dA$)

→ FOR DEFORMABLE CV, $\partial V/\partial t + \int_{CS} v\cdot dA = 0$

→ FOR STEADY FLOW, $\partial/\partial t(\cdot) = 0 \Rightarrow \int_{CS}\rho\vec{v}\cdot\vec{dA} = 0$ (USE THIS TO DEFINE MASS FLUX) (NOT NECESSARY ρ = CONST.)

EXAMPLE: THE FLUID IN DIRECT CONTACT WITH A STATIONARY SOLID BOUNDARY HAS ZERO VELOCITY. THUS, FLOW OVER A FLAT PLATE ADHERES TO THE PLATE SURFACE AND FORMS A BOUNDARY LAYER, AS DEPICTED IN FIGURE. THE VELOCITY AHEAD OF THE PLATE IS UNIFORM WITH $\vec{v} = u\hat{i}$, $U = 30$ m/s. WITHIN THE BOUNDARY LAYER ALONG cd, IT IS APPROXIMATED AS $u/U = 2(y/\delta) - (y/\delta)^2$. AT LOCATION d, $\delta = 5$ mm & $\rho = 1.24$ kg/m³. ASSUMING PLATE WIDTH INTO THE PLANE IS 0.6 m, CALCULATE THE MASS FLOW RATE ACROSS SURFACE bc OF CONTROL VOLUME abcd.

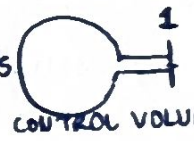


ρ = CONSTANT, NONDEFORMABLE CONTROL VOLUME $\Rightarrow \int_{ab} v\cdot dA + \int_{bc} v\cdot dA + \int_{cd} v\cdot dA + \int_{da} v\cdot dA = 0$ (0 AS $v = 0$; $v = 30\times 2\left(\frac{y}{\delta} - \frac{y^2}{\delta^2}\right)$)

$\Rightarrow -30\times 6\times\delta + \int_{bc} v\cdot dA + 0.6\int_{y=0}^{\delta}\left(2\frac{y}{\delta} - \left(\frac{y}{\delta}\right)^2\right)30\,dy = 0$

$\Rightarrow -180\delta + \int_{bc} v\cdot dA + 0.6\times 30\left\{\frac{2}{\delta}\cdot\frac{\delta^2}{2} - \frac{1}{\delta^2}\cdot\frac{\delta^3}{3}\right\} = 0$ $\Rightarrow -180\delta + \int_{bc} v\cdot dA + 6 = 0 \Rightarrow \dot{m}_{bc} = \rho\int_{bc}\vec{v}\cdot\vec{dA} = 1.24\times 6\times 0.005 = 0.0372$ kg/s

EXAMPLE: A TANK OF 0.05 m³ VOLUME CONTAINS AIR @ 1000 kPa AND 15°C. AT $t = 0$, AIR BEGINS ESCAPING THE TANK FROM A VALVE WITH AREA 80 mm². THE AIR PASSING THROUGH THE VALVE HAS A SPEED OF 250 m/s AND DENSITY OF 8 kg/m³. DETERMINE THE INSTANTANEOUS RATE OF CHANGE OF DENSITY IN THE TANK @ $t = 0$.



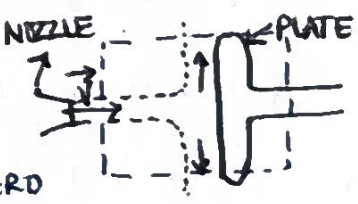
$\frac{\partial}{\partial t}\int\rho\,dV + \int_1\rho v\cdot dA = 0$ $\Rightarrow$ ASSUME: UNIFORM PROPERTIES IN TANK (VARY ONLY WITH TIME)

$\Rightarrow \frac{\partial}{\partial t}(\rho V) + 8\times 250\times 80\times 10^{-6} = 0 \Rightarrow \partial/\partial t(\rho V) = -0.16 \Rightarrow \frac{\partial\rho}{\partial t} = -3.2$ kg/m³/s

• $\vec{F}_B = M\vec{g}$ (TYPICALLY), $\vec{F}_S = \int_A -p\,\vec{dA}$ (NEGATIVE SIGN AS PRESSURE IS COMPRESSIVE)

• COMPONENT WISE FORM: $F_x = F_{Sx} + F_{Bx} = \frac{\partial}{\partial t}\int_{CV} u\rho\,dV + \sum_{CS} u\rho\vec{v}\cdot\vec{A}$ (FOR UNIFORM FLOW) → SAME FOR OTHER COMPONENTS!!

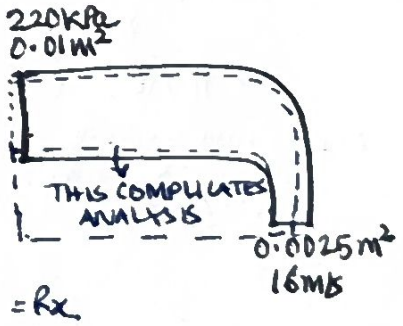
EXAMPLE: WATER FROM A STATIONARY NOZZLE STRIKES A FLAT PLATE AS SHOWN. THE WATER LEAVES THE NOZZLE @ 15 m/s; NOZZLE AREA IS 0.01 m². ASSUMING THE WATER IS DIRECTED NORMAL TO THE PLATE, AND FLOWS ALONG THE PLATE, DETERMINE THE HORIZONTAL FORCE YOU NEED TO HOLD IT IN PLACE.



x-COMPONENT $\Rightarrow F_{Sx} + F_{Bx} = \sum_{CS} u\rho\vec{V}\cdot\vec{A}$ → PRESSURE IS ATMOSPHERIC EVERYWHERE, NET ZERO → NO NET BODY FORCE IN THE x-DIRECTION.

$\Rightarrow R_x = 15\times 999\times 15\times 0.01 = 2.247$ kN

EXAMPLE: WATER FLOWS STEADILY THROUGH THE 90° REDUCING ELBOW SHOWN IN THE DIAGRAM. AT THE INLET TO THE ELBOW, THE ABSOLUTE PRESSURE IS 220 kPa AND CROSS-SECTIONAL AREA IS 0.01 m². AT THE OUTLET, CROSS SECTIONAL AREA IS 0.0025 m² AND VELOCITY IS 16 m/s. THE ELBOW DISCHARGES TO THE ATMOSPHERE. DETERMINE THE FORCE REQUIRED TO HOLD THE ELBOW IN PLACE.



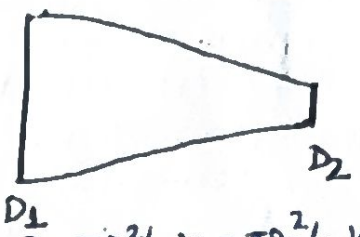
CONTINUITY EQN. $\Rightarrow 0.01\text{m}^2\times V_1 = 0.0025\text{m}^2\times 16\text{m/s} \Rightarrow v_1 = 4$ m/s (INLET VELOCITY)

ALONG x-DIRECTION, $-(220\text{kPa} - p_{atm})\times 0.01 \overset{\text{ga}}{=} -\sum v_1\cdot\rho\cdot v_1\times 0.01\text{m}^2$ (ONLY ONE SURFACE) $= R_x$

$\Rightarrow R_x = -1.19\times 10^3\times 0.01 - 999\times 0.01\times 16 = -1.35$ kN

ALONG y-DIRECTION (NEGLECT WEIGHT), $R_y = -999\times 16^2\times 0.0025 = -639$ N (NEGLECTING WEIGHT IS JUSTIFIED FOR A SMALL ELBOW, M IS SMALL)

EXAMPLE: WATER FLOWS STEADILY THROUGH A HORIZONTAL NOZZLE, DISCHARGING TO THE ATMOSPHERE. AT THE NOZZLE INLET THE DIAMETER IS D_1; AT THE NOZZLE OUTLET DIAMETER IS D_2. DERIVE AN EXPRESSION FOR THE MINIMUM GAGE PRESSURE REQUIRED AT THE NOZZLE INLET TO PRODUCE A GIVEN VOLUME FLOW RATE, Q. EVALUATE THE INLET GAGE PRESSURE IF $D_1 = 80mm$, $D_2 = 30\,mm$, AND THE DESIRED FLOW RATE IS $0.03\,m^3/s$.



ASSUMPTIONS → INCOMPRESSIBLE, INVISCID, STEADY, ALONG A STREAMLINE.

$p_1/\rho + V_1^2/2 + gz_1 = p_2/\rho + V_2^2/2 + gz_2$ (BERNOULLI), $\int \vec{V}\cdot d\vec{A} = 0 \Rightarrow D_1^2 V_1 = D_2^2 V_2$, WHERE $R = \pi D_1^2/4 \cdot V_1 = \pi D_2^2/4 \cdot V_2$

THUS, $p_1 = p_2 + \rho/2\,(V_2^2 - V_1^2) \Rightarrow p_1 - p_{ATM} = (p_2 - p_{ATM}) + \rho/2\left\{\left(\frac{D_1}{D_2}\right)^4 \cdot V_1^2 - V_1^2\right\} = \frac{\rho}{2}\left\{\left(\frac{D_1}{D_2}\right)^4 - 1\right\} \cdot \frac{Q^2 \cdot 16}{\pi^2 D_1^4}$

• WHEN WE HAVE A MOVING CONTROL VOLUME, THE VELOCITIES ARE MEASURED RELATIVE TO IT.

EXAMPLE: THE SKETCH SHOWS A VANE WITH A TURNING ANGLE OF 60°. THE VANE MOVES AT A CONSTANT SPEED, U = 10 m/s, AND RECEIVES A JET OF WATER THAT LEAVES A STATIONARY NOZZLE WITH SPEED V = 30 m/s. THE NOZZLE HAS AN EXIT AREA OF $0.003\,m^2$. DETERMINE THE FORCE COMPONENTS THAT ACT ON THE VANE.



CONTINUITY EQN. → $\rho_1 \cdot A_1 (30m/s - 10m/s) = \rho_2 A_2 V_2$ (ALONG THE NORMAL DIRECTION RELATIVE TO CV)

MOMENTUM EQN. x-DIRECTION → $R_x = \int_{A_1} -20^2 \times \rho_1 \times A_1 + \int_{A_2} (V_2 \cos 60°)\,\rho_2 \cdot A_2 \cdot V_2 = \rho A(-20^2 + V_2^2 \cos 60°)$

$= 999 \times 0.003\,(-400 + 1/2 \times 400)$

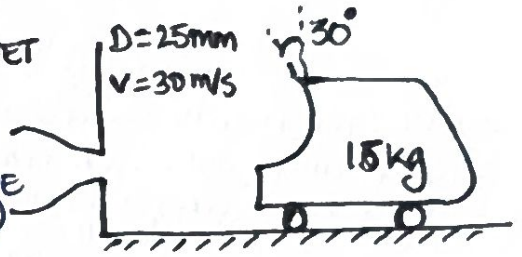
$= -599.4\ N$ { INERTIAL FRAME, SO THIS IS FINAL FORCE

MOMENTUM EQN. Y-DIRECTION → $R_y = \int_{A_2} (V_2 \sin 60°) \cdot \rho_2 A_2 V_2 = \rho A \cdot \sin 60° \cdot V_2^2 = 999 \times 0.003 \times \sqrt{3}/2 \times 400$

NEGLECTED WEIGHT OF WATER, VANE

$= 1.038\ kN$ { POSITIVE, SO THIS IS A LIFT FORCE.

EXAMPLE: THE CART IN THE FIGURE IS ACCELERATED ALONG A LEVEL TRACK BY A JET OF WATER THAT STRIKES THE CURVED VANE. DETERMINE THE TIME IT TAKES TO ACCELERATE THE CART FROM REST TO 10 m/s.



CONSERVATION OF MASS → $\sum \vec{V}\cdot d\vec{A} = 0 \Rightarrow -(30-v)A + AV_2 = 0 \Rightarrow V_2 = 30 - v$ (RELATIVE TO CV)

x-MOMENTUM → $\int_{CV} -\rho a\, d\forall = -(30-v)\rho(30-v)A + (-V_2 \sin\theta)\rho V_2 A$.

$\Rightarrow M\,dv/dt = -\rho A(30-v)^2 - \rho A \sin\theta\,(30-v)^2 = -\rho A(1+\sin\theta)(30-v)^2$

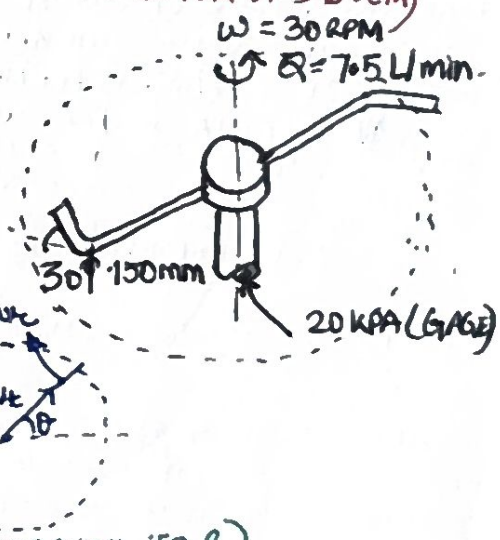
$\Rightarrow \int_0^{10} \frac{dv}{(30-v)^2} = \frac{\rho A}{M}(1+\sin\theta)\int_0^t dt \Rightarrow \frac{-1}{(30-v)}\Big|_0^{10} = \frac{\rho A}{M}(1+\sin\theta) \times t \Rightarrow t = 0.34\ sec.$

• REYNOLDS TRANSPORT THEOREM FOR ANGULAR MOMENTUM, $\left(\frac{dH}{dt}\right)_{SYSTEM} = \frac{\partial}{\partial t}\int_{CV} \vec{r}\times\vec{v}\,\rho\,d\forall + \int_{CS} \vec{r}\times\vec{v}\,\rho\,\vec{v}\cdot d\vec{A}$

(RATE OF CHANGE OF ANGULAR MOMENTUM OF SYSTEM)

ALSO, $\vec{T} = \vec{r}\times\vec{F}_S + \vec{T}_{SHAFT} + \int_{SYSTEM} \vec{r}\times\vec{g}\,dm$ (EXTERNAL TORQUE)

EXAMPLE: A SMALL LAWN SPRINKLER IS SHOWN IN THE SKETCH AT RIGHT. AT AN INLET GAGE PRESSURE OF 20 kPa, THE TOTAL VOLUME FLOW RATE OF WATER THROUGH THE SPRINKLER IS 7.5 LITRES PER MINUTE AND IT ROTATES AT 30 RPM. THE DIAMETER OF EACH JET IS 4mm. CALCULATE THE JET SPEED RELATIVE TO EACH SPRINKLER NOZZLE. EVALUATE THE FRICTION TORQUE AT THE SPRINKLER PIVOT.



CONSERVATION OF MASS → $Q/2 = \pi \times \frac{(4mm)^2}{4} \times V_R \Rightarrow V_R = 4.97\ m/s$

CONSERVATION OF ANGULAR MOMENTUM,

$\vec{v} = (V_t\cos\theta - \omega R\sin\theta)\hat{i} + (V_t\sin\theta + \omega R\cos\theta)\hat{j}$, $\vec{r} = R\cos\theta\,\hat{i} + R\sin\theta\,\hat{j}$

$\Rightarrow \vec{r}\times\vec{v} = \omega R^2\hat{k} \Rightarrow \frac{\partial}{\partial t}\int_{CV} \vec{r}\times\vec{v}\,\rho\,d\forall = \frac{\partial}{\partial t}\int_{\omega} (\omega R^2\hat{k})\cdot\rho\left(\frac{\pi D^2}{4}\right)x\,dx = 0$

ASSUMING INLET IS THIN, $\vec{r} = 0 \Rightarrow \vec{r}\times\vec{v} = 0$

AT OUTLETS, $\vec{r} = R\cos\theta\,\hat{i} + R\sin\theta\,\hat{j}$ (ASSUMING THAT THE ENDS ARE SMALL IN COMPARISON TO R)

$\vec{v} = (V_R\cos\alpha\sin\theta - \omega R\sin\theta)\hat{i} - (V_R\cos\alpha\cos\theta - \omega R\cos\theta)\hat{j} + V_R\sin\alpha\,\hat{k}$

$\Rightarrow \vec{r}\times\vec{v} = \hat{i}\cdot RV_R\sin\alpha\sin\theta - \hat{j}\,RV_R\sin\alpha\cos\theta - \hat{k}R(V_R\cos\alpha - \omega R)$

ACROSS TWO CONTROL SURFACES, θ VARIES BY π, ALSO WE KNOW $\sin(\pi+\theta) = -\sin\theta$, $\cos(\pi+\theta) = -\cos\theta$

$\Rightarrow \int_{CS}(\vec{r}\times\vec{v})\rho\vec{v}\cdot d\vec{A} = \rho Q/2\{\hat{i}\cdot RV_R\sin\alpha\sin\theta - \hat{j}\,RV_R\sin\alpha\cos\theta - \hat{k}R(V_R\cos\alpha - \omega R)\}$

$+\rho Q/2\{-\hat{i}\cdot RV_R\sin\alpha\sin\theta + \hat{j}\,RV_R\sin\alpha\cos\theta - \hat{k}R(V_R\cos\alpha - \omega R)\}$

$= -\rho QR(V_R\cos\alpha - \omega R)\hat{k}$

$\Rightarrow T_{SHAFT} = -\rho QR(V_R\cos\alpha - \omega R)\,\hat{k}$

$= -999 \times 7.5\,l/min \times 150mm \times (4.97 \times \cos 30° - 30\,RPM \times 150\,mm) = 0.0718\ Nm$